

Interpretable descriptors enable prediction of hydrogen-based superconductors at moderate pressures

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ABSTRACT

Room-temperature superconductivity remains elusive, and hydrogen-based compounds — despite remarkable transition temperatures (T_c) — typically require extreme pressures that hinder practical application. To accelerate discovery under moderate pressures, an interpretable framework based on symbolic regression is developed to predict T_c in hydrogen-based superconductors. A key descriptor is an integrated density of states (IDOS) within ± 1 eV of the Fermi level (E_F), which exhibits greater robustness than conventional single-point DOS features. The resulting analytic model links electronic-structure characteristics to superconducting performance (RMSE_{train} = 20.15 K, RMSE_{val} = 34.72 K) and is suitable for rapid materials screening. Guided by this model, four hydrogen-based candidates are identified and validated via calculation: Na₂GaCuH₆ with T_c = 42.04 K at ambient pressure (exceeding MgB₂), and NaCaH₁₂, NaSrH₁₂, and KSrH₁₂ with T_c up to 162.35 K, 86.32 K, and 55.13 K at 100 GPa, 25 GPa, and 25 GPa, respectively. Beyond rapid screening, the interpretable form clarifies how hydrogen-projected electronic weight near E_F and related features govern T_c in hydrides, offering a mechanism-aware route to stabilize high- T_c phases at reduced pressures.

1. Introduction

Hydrogen-based compounds provide a natural setting for conventional superconductivity, where light hydrogen modes and robust covalent frameworks jointly boost electron–phonon coupling and phonon energy scales, pushing superconducting transition temperatures (T_c) toward ambient temperatures under compression [1–3]. Within such frameworks, chemical “precompression” [2] embedded in the lattice lowers the metallization threshold of extended hydrogen networks, enabling superconducting phases at substantially reduced external pressure. Clathrate-like hydrides exemplify this design principle, achieving T_c values on the order of 200–250 K at megabar pressures, thereby positioning hydrogen-based materials as leading candidates for room-temperature superconductivity and underscoring the predictive power of computation-guided materials discovery [4–12].

Despite these advances, practical deployment is obstructed by the requirement of ultrahigh pressures for structural stability. Hydrides that retain high- T_c while remaining stable at moderate-to-low pressures are therefore a central target. Conventional first-principles workflows accurately capture electron–phonon interactions but are computation-

ally prohibitive for exhaustive exploration across vast chemical spaces, creating a discovery bottleneck precisely where rapid iteration is most needed [13].

Data-driven approaches offer a route to overcome this bottleneck while preserving physical insight. Instead of opaque black-box predictors [14], symbolic-regression models [15] can deliver compact, interpretable formulas that connect T_c to physically motivated descriptors—such as electron-localization measures, molecular/structural indices, hydrogen-projected density of states at the Fermi level (E_F) [16–20], and proxies for EPC strength and characteristic phonon scales. By revealing quantitative structure–property relations, such models enable fast prescreening and mechanism-aware optimization under pressure constraints. In this work, a physically constrained symbolic-regression framework is developed for hydrogen-based superconductors, producing an analytic T_c descriptor from readily computed features, validating its predictivity across known hydrides, and identifying chemically plausible candidates predicted to retain high- T_c under moderate pressures.

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2. Computational methods

All electronic structure calculations were performed within the framework of density functional theory (DFT) by the Vienna *Ab initio* Simulation Package (VASP) [21,22]. The exchange–correlation interactions among electrons were described using the Perdew–Burke–Ernzerhof (PBE) functional under the generalized gradient approximation (GGA) [23]. To ensure reliable results, especially under high pressures, a plane-wave cutoff energy of 1000 eV was employed. Brillouin zone integrations were performed using a Monkhorst–Pack k -point mesh with a reciprocal-space resolution of $2\pi \times 0.03 \text{ \AA}^{-1}$.

Phonon dispersions and the electron–phonon coupling (EPC) parameter λ were computed with the Quantum ESPRESSO (QE) [24–26] package using optimized norm-conserving Vanderbilt (ONCV) [27] pseudopotentials. The plane-wave kinetic-energy cutoff was 80 Ry. Self-consistent DFT calculations employed an $8 \times 8 \times 8$ k -grid as the coarse grid and a $16 \times 16 \times 16$ k -grid as the fine grid [28]. Density Functional Perturbation Theory (DFPT) [29] calculations used a $2 \times 2 \times 2$ phonon q grid.

For the Brillouin-zone integrations entering the electron–phonon coupling (EPC) calculation, the reported T_c values correspond to a double-delta broadening of $\sigma = 0.05$. Based on the Eliashberg spectral function $\alpha^2 F(\omega)$, the electron–phonon coupling constant λ and the logarithmic-average phonon frequency $\omega_{\log}$ were evaluated. The T_c was then estimated using the Allen–Dynes modified McMillan formula [30, 31] with a Coulomb pseudopotential of $\mu^* = 0.10$ throughout both training and application, the model is internally self-consistent and suitable for comparative ranking of candidate materials.

3. Result and discussion

The workflow for constructing and applying our T_c prediction model is illustrated in Fig. 1. We began with a database of hydrogen-based superconductors obtained from first-principles calculations, comprising 957 samples [19]. Data were cleaned based on two criteria: (1) exclusion of structures with imaginary phonon frequencies to guarantee dynamical stability; and (2) retention of only converged calculations yielding transition temperatures within the physically reasonable range of 0–500 K. This filtering process resulted in a training set of 343 samples. Electronic structure features were extracted using VASP and used as input variables for the sure independence screening and sparsifying operator (SISSO) [32]. To enhance the generalization capability of the model, we collected an external validation set comprising 72 samples collected from published literature [33–36]. This validation set was used to evaluate candidate models and guide the selection of the final prediction model. In the application stage, we employed a structure generator to produce large sets of hydride structures, applied the final prediction model to identify potential superconductor candidates. Subsequently, we carried out electron–phonon coupling calculations using QE to verify the candidates, ultimately identifying hydrogen-based superconductors with promisingly high transition temperatures. All electronic-structure descriptors used in this work were computed with consistent VASP settings. However, the T_c values for these sets were taken from the respective original references, where EPC calculations were performed with different computational parameters, introducing unavoidable inconsistencies. For the newly predicted superconductors, both the descriptors and T_c values were computed entirely in this work under identical settings.

Following the workflow outlined in Fig. 1 we obtained T_c and electronic features from 343 training samples after data cleaning. To assess the coverage and representativeness of our training set, we conducted statistical analysis on T_c , pressures, hydrogen content, and elemental distributions (Fig. S1). Although imbalances exist (such as higher proportions of low- T_c samples and hydrogen-rich systems), the dataset covers a broad physico-chemical space, providing a reliable foundation for subsequent model training.

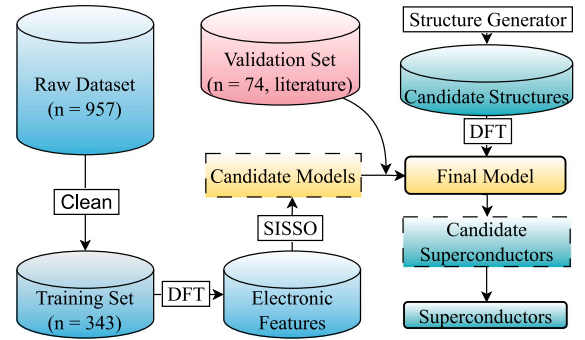


Fig. 1. Workflow for constructing and applying a prediction model of T_c in hydrogen-based superconductors using first-principles calculations and symbolic regression.

The relationships between key features and T_c are illustrated in Fig. 2. Pressure analysis (Fig. 2a) reveals that most of high- T_c superconductors occur in high-pressure regions, consistent with experimental observations of pressure-induced superconductivity where high pressure enhances electron–phonon coupling in hydrogen-based superconductors [3].

Regarding material composition (Fig. 2b–c), both hydrogen atomic fraction ($N_{\text{H/Tot}}$) and atomic number density ($\rho_{\text{at}} = N_{\text{tot}}/V$) show positive correlations with T_c . High- T_c superconductivity emerges predominantly when $N_{\text{H/Tot}} > 0.65$. This phenomenon can be attributed to the high vibrational frequency and light mass of hydrogen atoms, thereby enhancing superconducting pairing strength through the logarithmic average phonon frequency. Atomic number density (Fig. 2c) displays high correlation with T_c . This strong correlation can be understood from two perspectives: first, high atomic density implies tighter atomic packing and stronger interatomic interactions, favoring enhanced electron–phonon coupling constant λ ; second, high density typically corresponds to compressed states under high-pressure conditions, forming an intrinsic connection with the pressure dependence shown in Fig. 2a.

As shown in Fig. 2(d–f), we introduce the integrated DOS near the E_F (DOS^{Int}) as a key feature. All calculations perform integration within an energy window of ± 1 eV around the Fermi level. This choice is supported by a Kruskal–Wallis analysis comparing several integration windows, which reveals no statistically significant difference in the resulting DOS^{Int} values ($p = 0.99$; see Fig. S2). The ± 1 eV window is therefore adopted to ensure reproducibility. Compared to conventional approaches that only consider the DOS at the single point E_F (DOS^{E_F}), DOS^{Int} physically represents the number of active electrons near the Fermi surface, thereby offering clearer physical meaning. Furthermore, DOS^{Int} exhibits enhanced robustness compared to DOS^{E_F} under variations in computational parameters (e.g., k -point sampling, smearing schemes, and pseudopotential choices) as illustrated in Fig.S3(a–d), which can effectively reduce numerical instabilities. Additionally, standard pseudopotentials for H should be used when using this prediction model, as demonstrated in Fig.S3(e–f).

Since DOS^{Int} is an extensive quantity that scales with system size, we performed normalization to convert it into an intensive quantity, enabling meaningful comparisons across different system scales. We normalized by volume or hydrogen atom count to obtain physically meaningful descriptors. Fig. 2(d) shows the integrated hydrogen's s -orbital PDOS normalized by hydrogen atom count, which means the effective s -orbital electrons contributed per hydrogen atom ($\text{DOS}_{\text{H},s}^{\text{Int}}$), which exhibits a fan-shaped distribution with T_c , high- T_c superconductors tend to appear in the region with enhanced s -orbital participation, suggesting that hydrogen's s -orbital character is a favorable factor for superconductivity. Fig. 2(e) displays the effective p -orbital contributed electrons per hydrogen atom ($\text{DOS}_{\text{H},p}^{\text{Int}}$). It is evident that, in

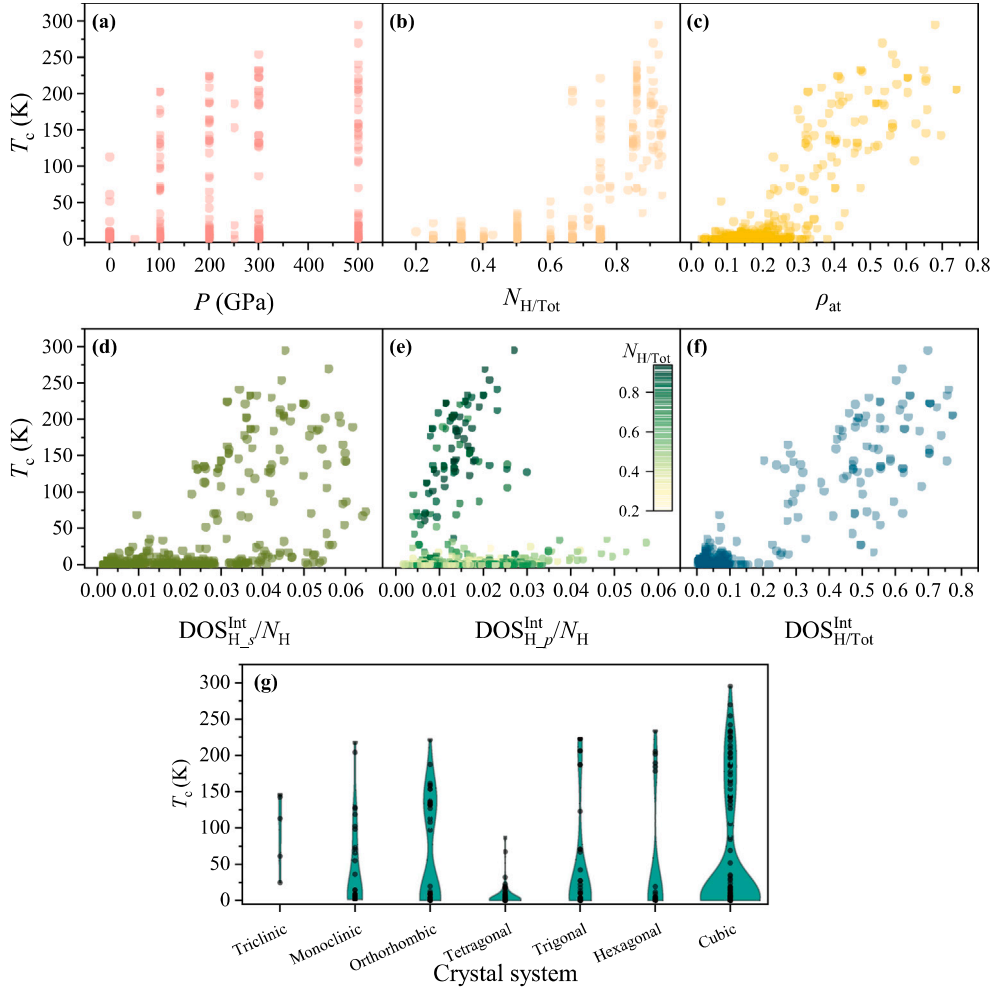


Fig. 2. Relationships between key parameters and T_c in the training set. (a) Pressure (P); (b) hydrogen atomic fraction ($N_{H/Tot}$); (c) atomic number density (ρ_{at}); (d) integrated hydrogen s -orbital partial density of states (PDOS) normalized by hydrogen atom count ($DOS_{H,s}^{Int}/N_H$); (e) integrated hydrogen p -orbital PDOS normalized by hydrogen atom count, color-coded by $N_{H/Tot}$ ($DOS_{H,p}^{Int}/N_H$), (f) hydrogen-to-total integrated DOS ratio ($DOS_{H/Tot}^{Int}$); (g) distribution of T_c across different crystal systems.

hydrogen-rich compounds, T_c exhibits a positive correlation with the contribution of hydrogen p -orbitals around the Fermi level, suggesting that the emergence of p -character in hydrogen plays an important role in enhancing superconductivity. Fig. 2(f) shows the hydrogen-to-total integrated DOS ratio ($DOS_{H/Tot}^{Int}$), representing the ratio of hydrogen effective electrons to total system effective electrons, suggesting a close relationship between the contribution of hydrogen's effective electrons to the total electron count with superconductivity.

Fig. 2(g) shows that T_c distributions vary significantly across different crystal systems. High- T_c structures are predominantly found in orthorhombic and cubic crystal systems, while other systems contain fewer superconductors with $T_c > 100$ K. Notably, the orthorhombic system exhibits a distinct bimodal distribution with peaks in both low-temperature (< 20 K) and high-temperature (100–200 K) regions, a clear gap is observed between them.

Using the SISSO symbolic regression approach, we trained the model on 343 samples, and generated a series of interpretable candidate predictive formulas. To prevent overfitting on the training set, we constructed an external validation set consisting of 72 hydrogen-based superconductors from recent theoretical studies in the literature. Each candidate prediction model was tested on this validation set, and the root mean square error (RMSE) was evaluated for both training and validation sets. Both errors were jointly used as the selection criterion. The final prediction model for the T_c of hydrogen-based

superconductors can be expressed as:

$$T_c^{pred} = a \times H_{act} + b \times D_p + c \quad (1)$$

where H_{act} is the hydrogen activity descriptor and D_p is the pressure-weighted dominance, defined as:

$$H_{act} = \sqrt{\frac{DOS_{H,p}^{Int}}{N_H}} \times DOS_{H/Tot}^{Int},$$

$$D_p = \sqrt[3]{\frac{DOS_H^{Int}}{\sum DOS_X^{Int}}} \times P$$

Here, $DOS_H^{Int}/\sum DOS_X^{Int}$ is the integrated DOS of non-hydrogen elements as a normalization factor for DOS_H^{Int} .

We obtained the coefficients: $a = 3930.62 \text{ K} \cdot (\text{atom}/\text{states})^{1/2}$, $b = -0.17 \text{ K/GPa}$, and $c = -4.69 \text{ K}$, it is evident that parameter H_{act} provides the dominant contribution to the T_c^{pred} , while parameter D_p has a much smaller weight, acting more as a fine-tuning term to T_c^{pred} .

In the construction of H_{act} , the factor $\sqrt{DOS_{H,p}^{Int}/N_H}$ serves as a per-atom p -orbital activity indicator. This term is obtained by normalizing the integrated p -orbital DOS of hydrogen per atom and taking its square root. When p -orbital character emerges at the Fermi surface, it indicates that the electronic system has departed from the

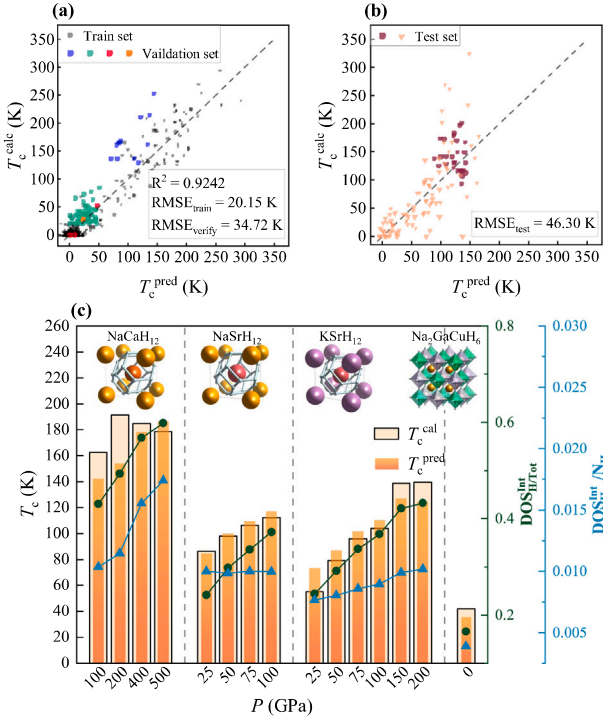


Fig. 3. Prediction and application results of the symbolic regression model for the T_c of hydrogen-based superconductors. (a) Linear fitting of predicted versus calculated values for the training set (gray dots) and the external validation set (colored dots); (b) Linear fitting of T_c prediction model on the external test set; (c) The four embedded images represent hydrogen-based superconductors newly found. Data for each superconductor are separated by dashed lines. Light yellow and orange bars indicate the calculated and predicted T_c of stable structures under different pressures, respectively. The green and blue symbols correspond to the quantities involved in the prediction formula, which are $\text{DOS}_{\text{H/Tot}}^{\text{int}}$ and $\text{DOS}_{\text{H}_p}^{\text{int}}/N_{\text{H}}$.

simple spherically symmetric ground state and entered a highly active hybridized state with enhanced directionality, spatial extension, and polarizability—essential prerequisites for strong electron-phonon coupling. This activation arises near the E_F can be through two mechanisms: (i) Under high pressure, the extension and overlap of hydrogen s - and p -orbitals lead to internal s - p hybridization; and (ii) in hydrides, the hydrogen $1s$ orbital hybridizes with the p -orbitals of neighboring atoms, thereby introducing additional p -character into the electronic states. The second factor, $\text{DOS}_{\text{H/Tot}}^{\text{int}}$, measures the concentration of active electrons of hydrogen near the E_F , is an important descriptor for superconductivity. This descriptor is physically meaningful, because in BCS-type superconductors, the density of states at E_F directly determines the strength of electron-phonon coupling and thus T_c . A higher $\text{DOS}_{\text{H/Tot}}^{\text{int}}$ indicates that hydrogen atoms, with their light mass and high phonon frequencies, dominate the electronic structure near the Fermi surface, creating favorable conditions for strong pairing interactions. Only when both factors are large enough does the system possess high-activity, high-concentration hydrogen electronic states near the E_F supporting elevated T_c .

As for parameter D_p , it introduces the normalized ratio between hydrogen and non-hydrogen integrated DOS, together with the external pressure. Its role is to capture how the hydrogen contribution is modified under different chemical environments. Pressure may introduce competing factors such as bandwidth broadening that slightly suppress T_c . Although its overall effect is weaker than that of H_{act} , it helps distinguish subtle variations in T_c among different materials.

The fitting performance of the prediction model is shown in Fig. 3(a). The model yields an RMSE of 20.15 K on the training

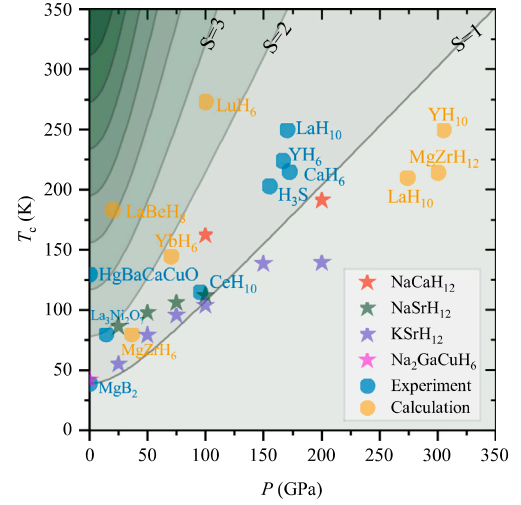


Fig. 4. Distribution of T_c for newly found superconductors under different pressures, as well as known experimental and theoretical superconductors. The background-colored contours represent iso-lines of the superconductivity index S .

set and 34.72 K on the validation set. Most data points lie close to the diagonal ($R^2 = 0.92$), indicating strong agreement between predicted and calculated values. This performance confirms the model's capability for efficient materials screening.

To further examine the generalization ability, we constructed an external test set from published works [16,37]. The test set consists of 141 samples, and the RMSE is 46.30 K, as shown in Fig. 3(b), most points remain close to the diagonal despite these differences, demonstrating that the predictions are consistent with reported values. Overall, this confirms that the symbolic regression model has strong generalization power and can predict T_c of complex hydrogen-based superconductors beyond the training domain.

The prediction model was further applied to identify new hydrogen-based superconductors using two sources of materials: (i) generating candidate structures using PyXtal [38] and optimizing them at 100 GPa, and (ii) retrieving “unexplored” structures from InvDesFlow [39] that had not been evaluated for superconductivity. Four promising compounds were obtained, NaCaH_{12} , NaSrH_{12} , KSrH_{12} and $\text{Na}_2\text{GaCuH}_6$. All four are confirmed to be dynamically stable under the corresponding pressures, as evidenced by phonon dispersion calculations showing no imaginary frequencies (see Fig. S6–S9 in Supporting Information). Among them, NaCaH_{12} , a derivative of CaH_6 [4,5], becomes dynamically stable at 100 GPa and exhibits $T_c = 162.35$ K with a superconductivity index [13] $S = 1.51$. NaSrH_{12} and KSrH_{12} , also CaH_6 derivatives, are stable at 25 GPa, with $T_c = 86.32$ K ($S = 1.86$) and 55.13 K ($S = 1.25$), respectively. $\text{Na}_2\text{GaCuH}_6$ exhibits a T_c of 42.04 K at ambient pressure, which was not recognized as a promising superconductor in the original study, surpassing the well-known superconductor MgB_2 [40]. Notably, this structure was subsequently also identified in the updated version of InvDesFlow [41], where electron-phonon coupling (EPC) calculations confirmed a similar T_c value of 42 K, corroborating our prediction and calculation. Superconducting performance of the newly identified materials, such as T_c , S and key descriptors for the four compounds across different pressure conditions, as summarized in Table S2.

As shown in Fig. 3(c) and Fig. S4, the predicted and calculated T_c values agree closely, confirming the reliability of the formula in screening hydrogen-based superconductors. For the newly found materials, the RMSE is 13 K, which indicates a promising prediction performance. Unlike the validation and test sets, we were able to control the computational parameters for the new materials to match

those used in the training set. As shown in Table S1, the present method achieves comparable accuracy with much higher computational efficiency. The electronic DOS for each material under different pressures are shown in Fig. S5. Crystal structures, phonon dispersions, and electron–phonon coupling properties of the four compounds are presented in Figs. S6–S9.

Fig. 4 illustrates that the newly predicted hydrogen-based superconductors in this study exhibit remarkably high S values, indicating superior overall performance. In particular, NaSrH_{12} reaches an S of 1.86 at 25 GPa. These findings suggest that such compounds may achieve appreciable T_c at medium-to-low pressures, or even ambient pressure, and thus motivate experimental synthesis and characterization of $\text{Na}_2\text{GaCuH}_6$ at ambient pressure and cage-like hydrides (NaSrH_{12} , KSrH_{12}) at about 25 GPa using diamond anvil cells, aiming to realize high- T_c superconductivity at moderate pressures.

4. Conclusion

In summary, this work demonstrates the power of combining physics-informed feature engineering with symbolic regression to advance the understanding and discovery of hydrogen-based superconductors. The integrated DOS descriptor (DOS^{Int}) not only addresses numerical instability in conventional calculations but also provides clear physical interpretation as the effective electron count near the Fermi surface. The model achieves satisfactory predictive performance across different datasets, validating its strong generalization capability.

We successfully identified four novel hydrogen-based superconductors by using the prediction model. All four materials exhibit superconductivity indices $S > 1$, confirming their superior overall performance.

These discoveries not only validate our physics-based interpretable machine learning approach but also establish rapid and viable new ways for achieving high- T_c superconductivity under milder pressure conditions. This physics-mechanism-driven data modeling framework greatly accelerates the discovery of future superconducting materials, thereby advancing the transition of hydrogen-based superconductors from theoretical prediction to practical application.

CRediT authorship contribution statement

Jiawei Chen: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Junhao Peng:** Writing – original draft, Project administration, Methodology, Conceptualization. **Yanwei Liang:** Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Renhai Wang:** Funding acquisition. **Huafeng Dong:** Writing – review & editing, Software, Resources, Project administration, Conceptualization. **Wei Zhang:** Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.mtphys.2026.102073>.

Data availability

Data will be made available on request.

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